

FACULTY OF COMPUTING**SE2030 - Software Engineering****LAB SHEET 05 - SEQUENCE DIAGRAM****Web-based E-Shopping Store**

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Group Type	Pro-rata / Individual Group
Project Topic	Web-based E-Shopping Store
Main Function	Place an Online Order

1. Selected Use Case Scenario

Use Case: Place an Online Order

Primary Actor	Customer
Preconditions	Customer can access the store and has selected products for purchase.
Success Result	Payment is accepted, an order record is created, inventory is updated and confirmation is displayed.
Supporting Components	Checkout UI, Order Controller, Cart/Inventory, Payment Service and Order Database

2. Main Flow

1. Customer proceeds to checkout after reviewing the cart.
2. Checkout UI sends the cart for validation.
3. Order Controller checks product availability through the Cart/Inventory component.
4. Available stock is reserved and the checkout total is returned.
5. Customer enters delivery information and confirms the order.
6. Order Controller requests payment processing from the Payment Service.
7. After successful payment, the Order Controller creates the order in the Order Database.
8. Inventory quantities are updated.
9. Order confirmation and order ID are returned to the customer.

3. Alternative Flows

- If stock is unavailable, the system informs the customer and allows the cart to be modified.
- If payment fails, the system returns a failure result and the customer may retry or cancel.
- If the customer cancels before successful payment, a confirmed order is not created.

4. Sequence Diagram

The sequence diagram uses boundary, control and entity-style participants to show how messages pass through the e-shopping system during checkout and order creation.

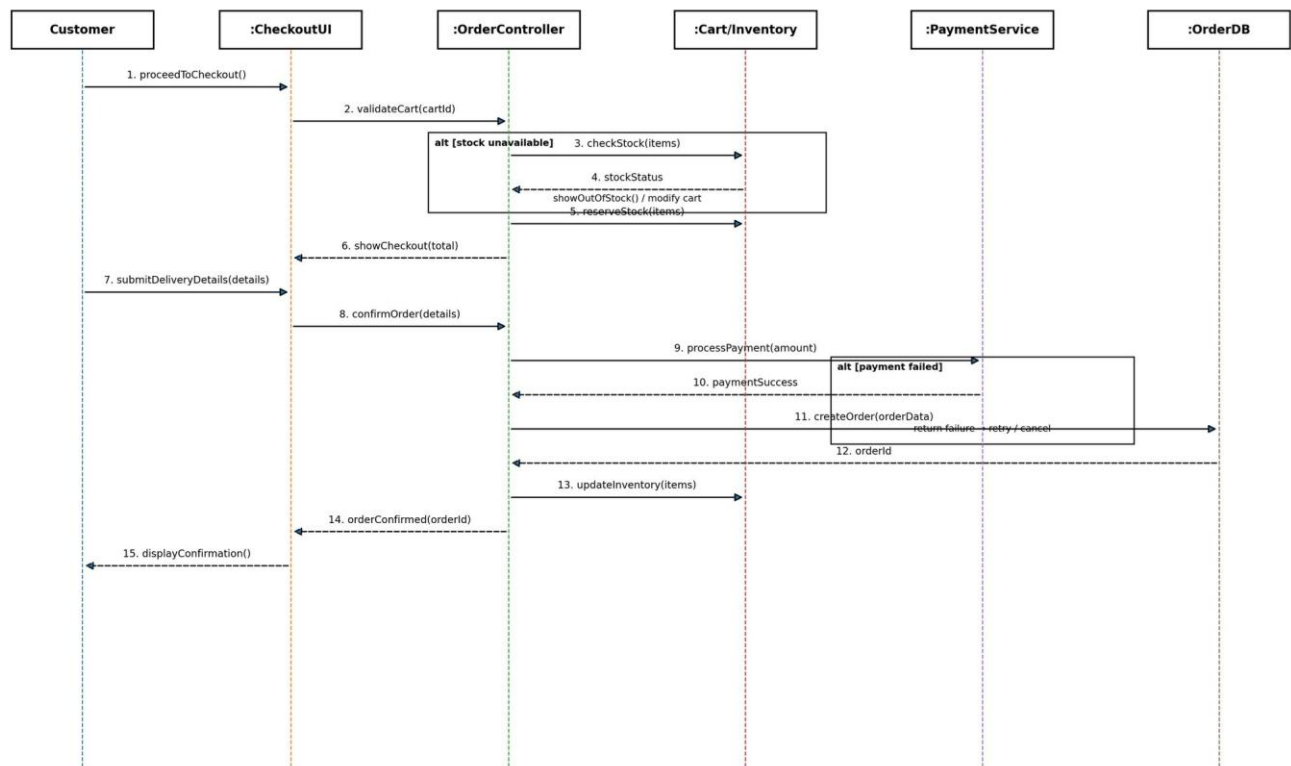


Figure 1: Sequence Diagram - Place an Online Order

5. Participant Responsibilities

Customer	Initiates checkout, supplies delivery information and receives confirmation.
CheckoutUI (Boundary)	Captures customer input and presents checkout/confirmation information.
OrderController (Control)	Coordinates validation, stock, payment and order creation.
Cart/Inventory (Entity)	Checks availability, reserves stock and updates quantities.
PaymentService	Processes the payment request and returns its status.
OrderDB (Entity)	Stores the confirmed order and returns the order identifier.

6. Conclusion

The diagram traces the complete interaction from checkout initiation to order confirmation. It also represents stockunavailable and payment-failure alternatives, keeping the sequence consistent with the use case scenario used in the previous lab.